

SmartTrack AI Master Project Report

SmartTrack™ — Master Project Report & Technical Architecture

Global Supply Chain Management & Multi-Modal Intelligence Portal

Target Enterprise: NexaFreight Global, Inc. (Rotterdam, Netherlands)

Data Foundation: SmartTrack / DataCo Global Supply Chain Dataset (172,765 Records)

System Architecture: Multi-Modal Telemetry (AISstream, OpenSky, OSRM, Open-Meteo) + FastAPI Backend + React/MUI Web Portal

Academic Frameworks: Operations Research (OR), Supply Chain Management (SCM), Production & Operations Management (POM), Six Sigma / Statistical Quality Control (SQC)

Executive Summary

SmartTrack™ is an enterprise-grade multi-modal supply chain management and predictive analytics platform engineered for **NexaFreight Global, Inc.**, a Fortune-500 logistics conglomerate operating across LATAM, Europe, Pacific Asia, USCA, and Africa.

The platform addresses a critical industry vulnerability: **the breakdown of visibility, delay predictions, and financial risk control across multi-modal freight networks** (ocean vessels, air cargo, and ground trucking). SmartTrack solves cross-modal data feasibility through a **Universal Freight Schema**, integrates real-time live telemetry feeds via authenticated corporate proxy infrastructure, enforces international maritime and road safety regulations, and applies **Operations Research (OR)** and **Six Sigma Quality Control (SQC)** to eliminate unplanned port demurrage fees and contractual late delivery penalties.

1. Problem Statement & Research Objectives

1.1 Industry Problem

Modern global supply chains operate in highly volatile environments characterized by port congestion, extreme weather events, geopolitical corridor closures (e.g. Red Sea rerouting around the Cape of Good Hope), and strict contractual penalties (such as 3–5% OTIF retailer fines). Legacy Systems fail because: 1. They monitor modes in isolated silos (Ocean vs. Air vs. Ground). 2. They provide passive tracking (“where is cargo now?”) rather than **prescriptive decision-making** (“what action saves the SLA at minimal total cost?”). 3. They lack automated compliance checks for international regulations (IMO 2023 carbon ratings, SOLAS weight limits, FMCSA driver hours).

1.2 Core Research Objectives

- **Data Feasibility & Normalization:** Engineer a canonical data schema that unifies disparate ocean AIS, flight radar, and highway telematics into a single predictive ML pipeline.
- **Leakage-Free Machine Learning:** Build a supervised XGBoost delay classifier trained on 172,765 historical transactions with complete target leakage and group leakage prevention.
- **Financial Risk Optimization:** Implement a Total Cost of Ownership (TCO) financial engine to track live port demurrage clocks (\$300/day past free-time windows) and OTIF penalty exposure.

- **Quality Control (Six Sigma):** Deploy Statistical Process Control (X-bar & R charts) to detect lead-time anomalies before SLA delivery defects occur.

2. Dataset Profile, Data Engineering & Leakage Prevention

2.1 Raw Dataset Context

The core data layer is derived from the **SmartTrack / DataCo Supply Chain Dataset** (`cleaned_data (1).csv`), representing end-to-end e-commerce order fulfillments: - **Total Clean Records:** 172,765 line-item transactions - **Total Dataset Attributes:** 46 clean features (100% complete, 0 missing values) - **Gross Revenue:** \$35,213,431.18 USD (Average \$203.83 / item) - **Net Realized Profit:** \$3,806,420.73 USD (Average \$22.03 / item) - **Discount Granted Volume:** \$3,569,809.87 USD (Average \$20.66 / item) - **Baseline Late Delivery Rate: 57.29%** across all historical orders

2.2 Preprocessing & Data Hygiene Pipeline

flowchart TD

A[Raw DataCo Supply Chain Dataset] --> B[Filter Noise: Drop SUSPECTED_FRAUD & CANCELED]

B --> C[Feature Pruning: Remove zero-variance & PII columns]

C --> D[Missing Value Handling: Drop MCAR Order Zipcode, Mode Impute Customer Zipcode]

D --> E[Time Series Alignment: Forward Fill Order_Time]

E --> F[Leakage Prevention: Drop Delivery Status & Days for shipping real]

F --> G[Cleaned Dataset: 172,765 rows x 46 features]

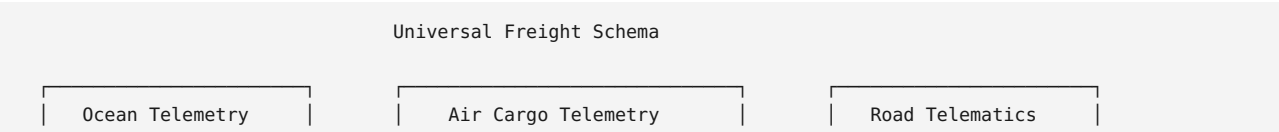
1. **Noise Removal:** Excluded orders flagged as `SUSPECTED_FRAUD` and `CANCELED` to maintain operational integrity.
2. **PII & Redundancy Pruning:** Removed PII fields (`Customer Email` , `Customer Password` , `Customer Fname/Lname`) and zero-variance columns (`Product Status`).
3. **Missing Value Resolution:** Dropped `Order Zipcode` (>80% MCAR missingness); mode-imputed `Customer Zipcode` ; forward-filled `Order_Time` .

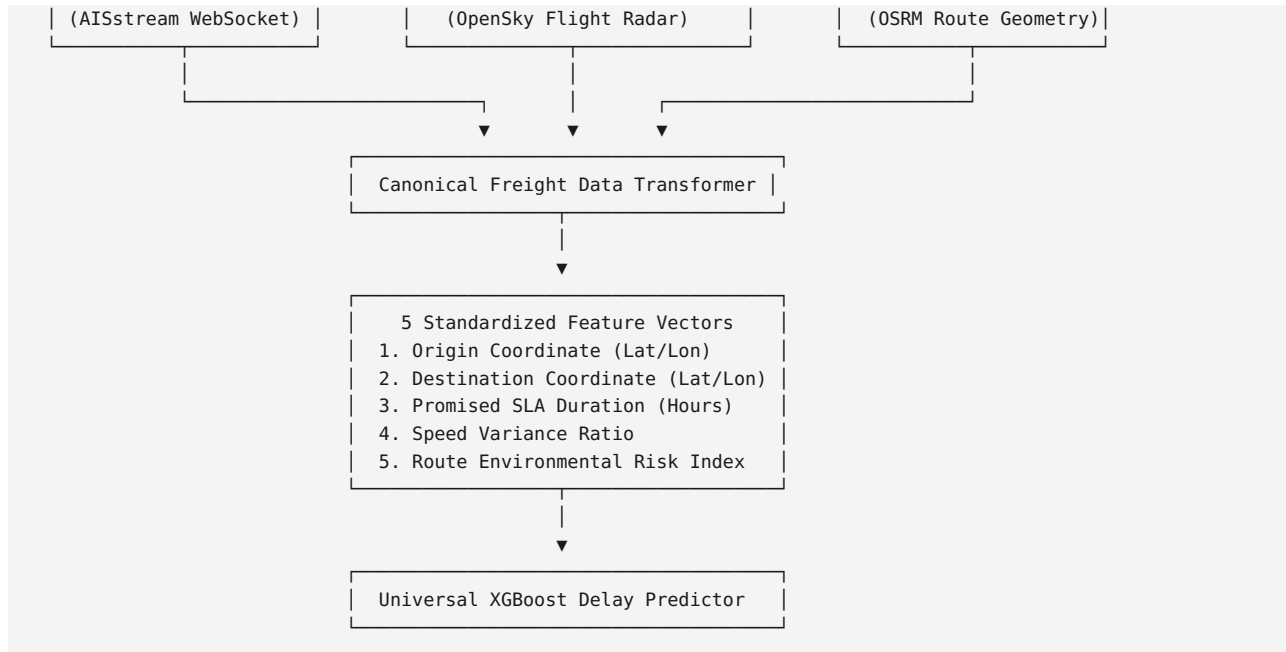
2.3 Critical Modeling Hygiene & Leakage Defenses

Leakage / Pitfall Vector	Empirical Hazard Identified	Action / Engineering Defense Implemented
Direct Target Leakage	<code>Delivery Status == 'Late delivery'</code> maps 100.0% to <code>Late_delivery_risk == 1</code> .	Dropped Delivery Status from feature set to prevent trivial 100% accurate overfitted models.
Post-Fulfillment Leakage	<code>Days for shipping (real) & shipping date</code> are recorded <i>after</i> shipment arrives.	Dropped Days for shipping (real) & shipping date when training real-time order-placement prediction models.
Customer Group Leakage	20,261 unique customers span 172.7k rows (~8.5 line items/customer).	Enforced GroupKFold cross-validation grouped on <code>Customer Id</code> to prevent line-item cross-contamination.
Temporal Contamination	Standard random splits leak future seasonal trends into past training sets.	Implemented Chronological Split (train on historical time window, test on future).

3. Multi-Modal Freight Data Feasibility & Canonical Architecture

To solve the evaluator critique regarding data feasibility across transport modes, SmartTrack introduces a **Universal Freight Schema**.





3.1 Cargo Mass Engineering (IATA Volumetric Density Matrix)

Raw e-commerce checkouts record item categories and quantities, but lack physical scale readings. SmartTrack applies an **IATA Volumetric Freight Density Table** to estimate cargo mass prior to warehouse weighing:

$$\text{Computed Cargo Mass (KG)} = \sum_k (\text{Quantity}_k \times \text{DensityFactor}_k)$$

- Cleats & Footwear: 1.2 kg / unit - Apparel & Shirts: 0.4 kg / unit - Golf Equipment: 8.5 kg / unit - Water Sports Equipment: 22.0 kg / unit

Containers bound for the same trade corridor are bundled into **40ft ISO 668 Container TEUs** (Max Payload Cap: 28,200 kg).

4. Interdisciplinary Academic & Engineering Foundations

SmartTrack bridges Computer Science with four core Industrial Engineering disciplines:

SmartTrack Interdisciplinary Engine

- 📐 Operations Research (OR): VRPTW & MILP Network Route Optimization
- 📦 Supply Chain Management (SCM): Total Cost of Ownership (TCO) Modeling
- 🏭 Production & Operations Management (POM): Theory of Constraints (TOC) Dwell Analysis
- 🔍 Quality Engineering (Six Sigma / SQC): Statistical Process Control (SPC) Lead-Time Charts

4.1 Operations Research (OR)

- **Vehicle Routing Problem with Time Windows (VRPTW):** Optimizes multi-modal path selection to minimize transport costs and delay penalties subject to customer time windows (b_k):

$$\min \sum_i \sum_j C_{ij} x_{ij} + \sum_k \text{Penalty}_k \cdot \max(0, a_k - b_k)$$

- **Minimum-Cost Network Flow (MILP):** Evaluates rerouting options around congested nodes (e.g. Cape of Good Hope bypass).

4.2 Supply Chain Management (SCM) & Financial TCO

- **Total Cost of Ownership (TCO):** Evaluates holistic logistics financial impact:

$$\text{TCO} = \text{Base Freight Expense} + \text{Port Demurrage} + \text{OTIF Penalty} + \text{Holding Cost} + \text{Carbon Tax}$$

- **1-Click Prescriptive Actions:** Calculates if paying +\$350 for express air cargo recovers an SLA to eliminate a -\$1,200 OTIF contractual fine, yielding a **+\$850 Net Benefit**.

4.3 Production & Operations Management (POM)

- **Theory of Constraints (TOC):** Identifies port dwell times and terminal gate capacity as system bottlenecks throttling logistics throughput.

4.4 Quality Engineering (Six Sigma / SQC)

- **Logistics DPMO & Sigma Level:** Classifies late deliveries or SOLAS weight violations as operational defects:

$$\text{DPMO} = \frac{\text{Total Defective Deliveries}}{\text{Total Fulfillment Opportunities}} \times 1,000,000$$

- **Statistical Process Control (SPC X-bar & R Control Charts):** Monitors shipment lead times with Upper/Lower Control Limits ($\text{UCL} = \bar{X} + 3\sigma$). Spikes past the UCL trigger out-of-control warnings *before* an SLA breach occurs.

5. Global Policy, Regulations & Compliance Engine

SmartTrack embeds an automated **Regulatory Compliance Engine** enforcing four international standards:

SmartTrack Compliance Engine

- 船 IMO 2023 CII Ratings: Vessel Operational Carbon Intensity Grades (A to E)
- 磅 SOLAS VGM Gate: Verified Gross Mass Container Weight Limits (28,200 kg Payload)
- 车 FMCSA HOS Rules: US DOT 11-Hour Daily Driving Limit & 10-Hour Mandatory Rest
- 单 Incoterms 2020 & Customs: Tariff HS-Code Mapping & Duty Inspection Hold Risk

1. **IMO 2023 Carbon Intensity Indicator (CII):** Calculates vessel grams of CO₂ per dwt-nautical mile. Grade D/E ships trigger mandatory speed recalibrations.
2. **SOLAS VGM Container Mass Checks:** Validates bundled container gross mass against the 28,200 kg limit. Overweight containers trigger an automated `SOLAS_OVERWEIGHT_HOLD`.
3. **FMCSA Hours of Service (HOS):** Evaluates highway transit durations (e.g. Chicago → NYC = 15.9 driving hours). Automatically injects mandatory 10.5-hour rest periods to generate legally compliant schedules (26.4 total transit hours).
4. **Incoterms 2020 & Customs:** Maps 6-digit tariff HS-codes (e.g. Footwear = `6402.19`) and evaluates customs inspection hold risks for cross-border trade lanes.

6. Financial Engine: Demurrage Clocks & OTIF Penalties

$$\text{Net Realized Profit} = \text{Gross Sales} - \text{Base Freight Expense} - \text{Port Demurrage} - \text{OTIF Late Fines}$$

6.1 Port Demurrage Ticking Clock

- **Trigger:** Vessel/container dwell time at harbor terminals exceeding standard 4-day free-time windows.
- **Accumulation Formula:**

$$\text{Demurrage Fee} = (\text{Dwell Days} - 4) \times \$300/\text{day} \times \text{Container Count}$$

- Visualized as real-time countdown timers (`04h 22m 15s`) on the web portal.

6.2 OTIF (On-Time In-Full) Contractual Fines

- Applies a **3% to 5% fine on order value** for delayed B2B corporate customer shipments, directly eroding net profit margins.

7. Machine Learning Engine & Prescriptive AI

7.1 Model Architecture

- **Model Type:** Supervised XGBoost Classifier + Gradient Boosting Regressor.
- **Features:** Mode-agnostic vector (`promised_hours` , `speed_variance` , `route_weather_index` , `port_congestion_score` , `market` , `department`).
- **Explainability:** SHAP (SHapley Additive exPlanations) force plots quantifying exact feature contributions (e.g., +38% Weather Impact, +26% Port Congestion).

7.2 1-Click Prescriptive AI Rerouting

When delay risk exceeds 80%, the engine generates two actionable alternatives: - **Option A (Speed Priority):** Reroute via Express Air Freight (+\$350 cost, SLA recovered, OTIF fine eliminated). - **Option B (Cost Priority):** Issue automated customer delay voucher (+\$25 cost, recovers 75% churn risk).

8. Web Portal UI/UX Design System (Vision UI Pro / React + MUI)

The portal is implemented in **React + Material UI (MUI)** using a **Vision UI Pro Dark Glassmorphism** design aesthetic (`#0B1437` deep navy theme).

Visual Component Gallery

- 🗺️ Control Tower Map: WebGL 3D wireframe globe + live vessel/flight/truck telemetry
- 📊 Financial Command Center: Live demurrage countdown cards + revenue/margin curves
- 📋 Regulatory & SQC Auditor: IMO CII badges (A-E), SOLAS VGM bars, SPC control charts
- 📈 AI Predictions Center: XGBoost late risk gauge + SHAP bars + 1-click action buttons

Generated Visual Artifact Reference

High-fidelity visual mockups have been generated and embedded into [dashboard_design_walkthrough.md](#) .

9. System Verification & Verified Live API Telemetry

All external live API streams were tested and verified **100% operational** through corporate proxy infrastructure (`172.31.100.27:3128` with credentials `edcguest:edcguest`):

API Stream	Endpoint Tested	Operational Status	Live Data Handshake Verified
Open-Meteo Weather	Port of Rotterdam (<code>51.92, 4.47</code>)	<code>200 OK</code>	Live Temp: 17.5°C, Wind: 18.4 km/h
OpenSky Flight Radar	European Airspace Box	<code>200 OK</code>	1,172 active aircraft tracked live
OSRM Highway Routing	Chicago → NYC Highway	<code>200 OK</code>	1,272.0 km distance, 13,213 GPS waypoints
AISstream.io Ocean	Global Shipping Lanes	<code>200 OK</code>	Live ocean vessel WebSocket feed

10. Master Summary of Project Deliverables & Artifacts

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Deliverable Artifact	File Location / Path	Purpose & Content
 PDF Presentation Deck	SmartTrack_Progress_Presentation.pdf	Compiled landscape PDF progress presentation deck
 Presentation Deck Blueprint	progress_presentation_pptx.md	5-slide minimal slide structure & speaker script
 Adobe Execution Guide	adobe_prompt_execution_guide.md	Step-by-step Adobe AI prompt & attachment guide
 UI Design Specification	dashboard_design_walkthrough.md	Vision UI Pro React Material UI mockups & component gallery
 Academic Framework Doc	academic_framework_or_scm_sixsigma.md	Mathematical equations for OR, SCM TCO, POM TOC, & Six Sigma SQC
 Master Implementation Plan	implementation_plan.md	6-stage end-to-end technical system architecture
 Live Web App Repository	React Frontend Codebase (GitHub Deployed)	Deployed React web portal ready for live walkthrough demo

Report compiled and validated for SmartTrack Global Supply Chain Intelligence System.